Theoretische Elektrotechnik
Universität Paderborn

Python Fundamentals

submission deadline: 20.10.2026 11:00AM (20 points)

Create a python script file (.py) with the following python commands:

1. **Vector and matrix manipulations (7P):**

   (a) Get help on function"numpy.zeros".

   (b) Get help on function "numpy.ones".

   (c) Get help on function "numpy.random.rand".

   (d) Define 1×5 (1-row and 5-columns) vector a with all elements equal to 0.

   (e) Define 3×1 (3-rows and 1-column) vector a with all elements equal to 1.

   (f) Define 1×5 vector a with elements equal to $1, 2, 3, 4, \pi$, correspondingly.

   (g) Define 3×1 vector a with elements equal to $0.1, 0.2, 0.3$, correspondingly.

   (h) Define 1-row vector a with the elements' values from 3 to 27 with step 3 by using *numpy.arange* or *range*.

   (i) Define 1-row vector a of 10 linearly equally spaced points between x_1 and x_2, where x_1 and x_2 are the first and last numbers of your student id (use *numpy.linspace* function).

   (j) Define 3×3 matrix a of random number elements.

   (k) Define 3×2 matrix $a = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$.

   (l) Check the size of the matrix a.

   (m) Define 3×3 matrix $a = \begin{pmatrix} 1 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{pmatrix}$.

   (n) Transpose the matrix a.

   (o) Show (print on display) the element a_{21} of the transposed matrix a.

   (p) Show (print on display) the second row of the transposed matrix a.

   (q) Show (print on display) the third column of the transposed matrix a.

   (r) Define matrix $a = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ and $b = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$.

   (s) Sum matrices a and b, $c = a + b$.

   (t) Multiply matrices a and b (matrix product).

   (u) Multiply matrices a and b element-by-element, i.e. $c_{ij} = a_{ij} * b_{ij}$.

   (v) Solve the linear system of equations $ax = b$ by using *inv()* function, if $a = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$ and $b = \begin{pmatrix} 8 \\ 10 \\ 12 \end{pmatrix}$.

   (w) Solve the same system by using operator *numpy.linalg.solve*.

2. **Data analysis functions (4P):**
   (a) Calculate the square root of your student id.
   (b) Define the 1-row vector $a(11, 7, 32, 34, 13, your_student_id_number)$.
   (c) Get the value of the last element of vector a .
   (d) Get the part of array a starting from 2nd element using slicing.
   (e) Find minimum value of the vector a.
   (f) Find maximum value of the vector a.
   (g) Calculate the mean value of the vector a.
   (h) Make summation of elements of the vector a.
   (i) Calculate the square roots of all elements of the vector a.
   (j) Sort the vector a in the ascending order a.
   (k) Find the roots of the polynomial equation $x^4 + 4x^3 - 5x^2 + 6x - 9 = 0$.

3. **Making complex numbers (1P):**
   (a) Make a complex number $a = 2 + 3 * j$.
   (b) Get the real part of a.
   (c) Get the imaginary part of a.
   (d) Get the complex conjugate of a.
   (e) Get the value of complex number i.

4. **Plotting Tools (2P):**
   (a) Get help on function "matplotlib.pyplot.plot".
   (b) Plot the graph of $y = exp(x)$, for $-2\pi \leqslant x \leqslant 2\pi$.
   (c) Plot the graph of $y = log(x)$, for $0 < x \leqslant 2\pi$.

5. **Python functions (6P):**
   Write the Python function (.py) that (a) creates the square matrix a of the size $n \times n$ with all diagonal elements equal to 3 and all other elements equal to 1 (you may use *diag* and *ones* functions) and vector b of size n with elements $b_i = \sum_{j=1}^{n} j * a_{ij}$, b) solve the linear system of equations $ax = b$. The function takes one input parameter - n, and output is the solution vector - x. Run the code for n equal to last number of your student id, if this number is equal to 0 or 1 use the number 9, etc. Copy the obtained values of the vector x at the end of this file.

Hints and Notes:
In this exercise, you are required to use:

- NumPy for vectors and matrices
- cmath for complex number operations

Add the following lines at the top of your code:

- import numpy as np
- import cmath

Instructions for Submission:

For homework submission, please submit your solutions in both Python script (`.py`) and Jupyter Notebook (`.ipynb`) formats.

1. Submit your solutions to Exercises 1–5.
2. For Exercise 5, include the required function and ensure that it can be executed correctly.
3. You may additionally submit an HTML export (`.html`) showing your code, results, and plots.

Please ensure that all code runs correctly and that the required numerical results and figures are included.